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E-COMMERCE PLATFORM FOR RESPONSIBLE CONSUMPTION AND DIGITAL COMMERCE

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements for
the Course of*

CSA1110 - OBJECT ORIENTED ANALYSIS AND DESIGN

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE ENGINEERING

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**E-COMMERCE PLATFORM FOR RESPONSIBLE CONSUMPTION AND DIGITAL COMMERCE**” has been carried out by **Sayed Fazal (192311291)**, **Shaik Fazul (192411063)**, **Shivaram Sagar (192373054)**, **Nallabrolu Murali Sai (192372309)**, **Venkata Sai Giresh (192372278)** under the supervision of **Dr. A. Manimaran** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, **Sayed Fazal (192311291), Shaik Fazul (192411063), Shivaram Sagar (192373054), Nallabrolu Murali Sai (192372309), Venkata Sai Gireesh (192372278)** of the **Department of CSE**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled **“E-COMMERCE PLATFORM FOR RESPONSIBLE CONSUMPTION AND DIGITAL COMMERCE”** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement
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Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Venkata Sai Giresh (192372278)	Analyzed project requirements and identified stakeholders, actors, and major functionalities.	Prepared SRS and initial use case model.	Reviewed requirements and verified major system interactions.	Prepared requirement analysis, existing system, objectives, and problem statement.	20%
Shivaram Sagar (192373054)	Designed the overall system structure using OOAD principles.	Created UML diagrams including Class, Sequence, Activity, Component, and Deployment diagrams.	Verified class relationships, interactions, and system design models.	Prepared system design, architecture, and UML documentation.	20%
Nallabrolu Murali Sai (192372309)	Managed customer, seller, and product management functionalities.	Implemented authentication, product catalog, search, shopping cart, and sustainable product	Tested registration, login, product operations, search, cart, and recommendations.	Prepared module description, implementation details, UML diagrams, and results.	20%

		features.			
Sayed Fazal (192311291)	Managed order creation, validation, status management, and order–payment interaction.	Designed and implemented the Order Processing and Digital Commerce module using OOAD.	Tested order creation, validation, payment interaction, and status workflows.	Prepared order processing, implementation, UML diagrams, and testing documentation.	20%
Shaik Fazul (192411063)	Managed administration, system testing, and deployment activities.	Implemented admin functions for users, products, orders, inventory, and platform management.	Performed unit, integration, system, and performance testing.	Prepared testing results, administration, deployment, conclusion, limitations, and future scope.	20%
Total					100%

ABSTRACT

Conventional e-commerce platforms are typically optimised for transaction volume and convenience, with limited built-in support for helping consumers make more responsible purchasing decisions or for giving businesses structured visibility into order-level sustainability considerations. At the same time, an e-commerce platform is itself a substantial software-engineering undertaking: it must reliably manage user accounts, product catalogues, shopping carts, order processing, payment interaction, and order tracking as a coherent, well-structured object-oriented system rather than an ad-hoc collection of scripts. This report presents the design and implementation of an E-Commerce Platform for Responsible Consumption and Digital Commerce, developed using Object-Oriented Analysis and Design (OOAD) methodology and structured around five interacting modules covering user management, product cataloguing, order processing, payment interaction, and order tracking.

As an individual contribution within this five-member team project, the author's assigned module is Order Processing and Digital Commerce, covering the order lifecycle from cart-to-order conversion through order creation, order validation, order status management, and order–payment interaction, modelled using UML class and sequence diagrams consistent with standard OOAD practice. The methodology followed a structured OOAD process: requirement definition, comparative evaluation of alternative architectural approaches, class- and sequence-level design of the Order Processing module and its interfaces with the Shopping Cart and Payment modules, and verification of the module's order-creation, validation, and status-transition behaviour against a defined test plan.

Keywords: Object-Oriented Analysis and Design; E-Commerce; Order Processing; UML; Digital Commerce; Responsible Consumption

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
OOAD	Object-Oriented Analysis and Design	—
UML	Unified Modeling Language	—
UI	User Interface	—
DB	Database	—
SDG	Sustainable Development Goal	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

E-commerce platforms have become the primary channel through which many consumers discover, evaluate, and purchase products, and their internal software architecture directly shapes both the reliability of the purchasing experience and the platform's capacity to support more deliberate, responsible consumption behaviour. The E-Commerce Platform for Responsible Consumption and Digital Commerce is structured as an object-oriented system comprising user management, product catalogue, shopping cart, order processing, payment interaction, and order tracking, designed using Object-Oriented Analysis and Design (OOAD) methodology so that each concern is represented by well-defined classes, responsibilities, and interfaces rather than an undifferentiated codebase.

Within this overall platform, the Order Processing and Digital Commerce module, the author's individual contribution, is responsible for converting a customer's shopping cart into a validated order, managing the order's status through its lifecycle, and coordinating with the payment module to confirm the order. This module sits at a structurally important point in the platform: it depends on accurate cart and product data from upstream modules, and its output (a confirmed, correctly priced order) is what the payment and order-tracking modules subsequently act on, making the correctness of its OOAD design and implementation significant to the reliability of the platform as a whole.

1.2 Need for the Project

User and product management is a critical part of an e-commerce platform because incorrect user information, poor product management, or inconsistent data can affect the shopping experience and other modules. Unstructured management of users and products can make the system difficult to maintain, test, and extend. Therefore, applying OOAD methodology provides a clear structure for managing users, sellers, products, categories, and reviews. A well-designed module also provides a stable foundation for features supporting responsible consumption, such as eco-friendly, recyclable, and reusable product information.

1.3 Problem Statement

The engineering problem addressed by this module is the object-oriented design and implementation of a User and Product Management component that reliably manages user registration, authentication, profiles, sellers, and product information. The module must support product creation, updating, deletion, searching, and categorization while maintaining accurate database records and providing responsible-consumption attributes such as eco-friendly, recyclable, and reusable. It must also integrate correctly with other modules developed independently by team members.

1.4 Project Aim

To design, implement, and test a User and Product Management module, using Object-Oriented Analysis and Design methodology, that reliably manages users, sellers, and product information within the broader E-Commerce Platform for Responsible Consumption and Digital Commerce.

1.5 Project Objectives

- To design the overall e-commerce platform architecture and identify the Order Processing module's place within it.
- To model the Order Processing and Digital Commerce module using UML class and sequence diagrams.
- To implement order creation, order validation, and order status management functionality.
- To implement the module's interface with the Payment module for order confirmation.
- To test the order lifecycle (creation, validation, confirmation, cancellation) against a defined test plan.
- To evaluate the Order Processing module's design against the project's functional and non-functional requirements.

1.6 Scope

Included:

- OOAD modelling (class diagram, sequence diagram) of the User and Product Management module.

- User registration, login/authentication, profile management, and logout.
- Seller management, including adding, updating, deleting, and viewing products.
- Product catalogue management, including product details, categories, search, and filtering.
- Product reviews and ratings, along with responsible-consumption attributes such as eco-friendly, recyclable, and reusable.

Excluded:

- Detailed design and implementation of the Shopping Cart, Order Processing, Payment, and Order Tracking modules, which are the responsibility of other team members.
- Advanced payment gateway integration and transaction processing.
- AI-based product recommendations and advanced sustainability calculations.
- Features not actually implemented in the User and Product Management module.

Assumptions:

- Users provide valid and well-formed information during registration and profile updates.
- Sellers provide accurate product details, prices, quantities, categories, and responsible-consumption attributes.
- The MySQL database is available and accessible for storing user and product information.
- Other modules can access the required user and product information through the defined module interfaces.

Boundary Conditions:

- The module's responsibility ends at user/profile management and product catalogue management; cart, order processing, payment, and delivery operations are outside its scope.
- User access is limited to the roles and authentication mechanisms defined for the system.
- Product operations are restricted to the functions supported by the module, such as add, update, delete, search, filter, and view.
- Concurrent data handling and advanced security mechanisms are outside the current scope unless specifically implemented.

1.7 Expected Engineering Outcome

The expected outcome is a working User and Product Management module, designed using OOAD methodology, that correctly manages user registration, authentication, profile information, seller operations, and product data. The module is expected to support product addition, updating, deletion, searching, category filtering, and responsible-consumption attributes such as eco-friendly, recyclable, and reusable, while providing reliable data exchange with the other modules of the e-commerce platform.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature was identified covering Object-Oriented Analysis and Design and UML modelling practice for order-processing and e-commerce systems, the formal UML specification itself, and recent research on sustainability and responsible consumption in e-commerce, since these two threads (OOAD methodology and responsible-consumption e-commerce) together define the technical and domain context of this project. Sources were selected for direct relevance to the Order Processing module's design approach and to the platform's responsible-consumption objective, and are summarised in Table 2.1.

2.2 Review of Existing Work

Table 2.1 — Literature Review Summary

Title	Author(s) & Year	Focus	Key Findings
Mastering Object-Oriented Design: A Practical Case Study in Order Processing Systems Using UML Class Diagrams	Visual Paradigm, 2026	UML class-diagram case study for an order processing system	Demonstrates how class diagrams bridge business requirements and technical implementation for order management, improving maintainability
UML Distilled: A Brief Guide to the Standard Object Modeling Language	M. Fowler, 3rd ed., 2003	OOAD/UML methodology reference	Establishes standard notation and practice for class, sequence, and use-case modelling used throughout this project
Unified Modeling Language (UML), Version 2.5.1	Object Management Group, 2017	Formal UML specification	Defines the authoritative notation for class and sequence diagrams (e.g., multiplicities, message syntax)

Title	Author(s) & Year	Focus	Key Findings
			adopted in Chapter 3
E-commerce ecosystems as catalysts for sustainability: A multi-case analysis	T. Wulfert, R. Woroch, G. Strobel, T. Schoormann, and L. Banh, 2024	Multi-case study of sustainability initiatives across e-commerce ecosystems	Finds that e-commerce ecosystem actors address SDG 12 (Responsible Consumption and Production) and identifies dependencies between sustainability goals within e-commerce initiatives
SEER: Sustainable E-commerce with Environmental-impact Rating	M. S. Islam et al., 2022	Concept design and controlled experiment on eco-impact ratings in online shopping	Found that displaying explainable environmental-impact information during online shopping produced significantly more eco-friendly consumption behaviour

The Visual Paradigm case study and Fowler's UML Distilled together establish the class-diagram-centred modelling approach adopted for the Order Processing module in this report, with the Object Management Group's formal UML specification providing the authoritative notation (multiplicities, message syntax) used in Section 3.5 [1], [2], [3]. On the responsible-consumption side, Wulfert et al.'s multi-case analysis of e-commerce ecosystems found that platform actors address SDG 12 (Responsible Consumption and Production) through a range of sustainability initiatives, while identifying data-integration and cross-actor coordination as recurring challenges [4]. Islam et al.'s SEER study provides direct experimental evidence that redesigning an e-commerce platform to surface environmental-impact information at the point of purchase can measurably shift consumer behaviour toward more sustainable choices [5], which supports the design rationale for a platform architecture (Section 3.5) that keeps the Order Processing module's interfaces open enough for such consumption-

facing features to be added by other modules without requiring the order-handling core to be redesigned.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Ad-hoc / script-based order handling	Fast to build initially	Difficult to test, extend, or reason about; error-prone as complexity grows	Used as the baseline against which the OOAD approach is evaluated
UML class-diagram-modelled order processing [1], [2], [3]	Clear, testable contract between order handling and other modules; supported by standard notation	Requires upfront modelling effort before implementation	Directly adopted as the design approach for the Order Processing module (Section 3.5)
Sustainability-integrated e-commerce ecosystems [4]	Demonstrated real-world sustainability initiatives across e-commerce actors	Requires cross-actor coordination and data integration, which is complex at ecosystem scale	Informs the platform's responsible-consumption objective at the architecture level (Figure 3.1), outside the Order Processing module's direct scope
Environmental-impact rating at point of purchase [5]	Experimentally shown to shift consumer behaviour toward sustainability	Concept-stage design; requires reliable environmental-impact data generation	Identified as a candidate future extension (Section 5.2) building on this module's stable order interface

2.4 Research / Engineering Gap

The reviewed literature shows that UML/OOAD-based user and product management is well established as a software-engineering practice [1], [2], [3], while sustainability-oriented e-commerce research highlights the importance of providing customers with information about environmentally responsible products and supporting responsible purchasing behaviour [4], [5]. However, the reviewed work does not clearly

specify how user management, seller operations, product catalogue management, and responsible-consumption attributes can be integrated within a single OOAD-based module. This creates scope for a project that treats User and Product Management as a cleanly modelled and independently testable component, providing a stable foundation for other modules such as Shopping Cart and Order Processing.

2.5 Proposed Contribution

This project's contribution is an OOAD-modelled User and Product Management module, with class and sequence diagrams following standard UML practice [1], [2], [3]. The module provides a structured interface for user registration, login, profile management, product addition, updating, deletion, searching, category filtering, and product reviews. It also incorporates responsible-consumption attributes such as eco-friendly, recyclable, and reusable products, supporting the platform's broader sustainability objective informed by [4] and [5]. The module is designed to interact with other platform components without requiring major changes to its core user and product management functions.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Customer	Easy registration, login, profile management, and product search.
Seller	Ability to add, update, delete, and manage product information.
Project Team (Other Modules)	Stable user and product data that can be accessed by other modules.
Faculty / Evaluator	Clearly tested and documented User and Product Management module.
Database / System	Accurate and consistent storage of user, seller, and product information.

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
User registration and login	Functional	Valid users can successfully register and log in
Manage user profiles	Functional	Profile details can be updated and saved correctly
Manage products	Functional	Sellers can add, update, delete, and view products
Search and filter products	Functional	Relevant products are displayed based on search/category
Support responsible-consumption attributes	Functional	Eco-friendly, recyclable, and reusable details are stored and displayed
Stable module interface	Non-Functional	User and product data can be accessed by other modules

3.2 Constraints & Applicable Considerations

No specific regulatory standard governs the internal object-oriented design of this academic User and Product Management module. The main technical consideration is the use of standard UML notation defined by the Object Management Group (OMG), which guides the interpretation of the class and sequence diagrams, including relationships, multiplicities, and message interactions [3]. The platform's broader Responsible Consumption and Production objective is aligned with UN Sustainable

Development Goal 12 [4], which provides a domain reference for including responsible-consumption attributes such as eco-friendly, recyclable, and reusable products.

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
User registration	Valid users can create accounts	—	High
User authentication	Valid credentials allow secure login	—	High
Profile management	User details can be updated correctly	—	Medium
Product management	Add, update, delete, and view products	Operations	High
Product search	Relevant products displayed for search	—	High
Category filtering	Products filtered by category	—	Medium
Responsible-consumption attributes	Eco-friendly, recyclable, and reusable details supported	Attributes	Medium
Interface stability	No breaking changes after module integration	—	High
Category Filtering	Products filtered by category	---	Medium
User registration	Valid users can create accounts	—	High

3.4 Alternative Solutions & Evaluation

Alternative 1 — Ad-hoc user and product management: User and product operations are implemented using loosely organized functions without clear object-oriented structure.

Alternative 2 — OOAD-modelled User and Product Management module (selected): User, Seller, Product, Category, and ProductReview are represented as classes with defined relationships and operations using UML diagrams.

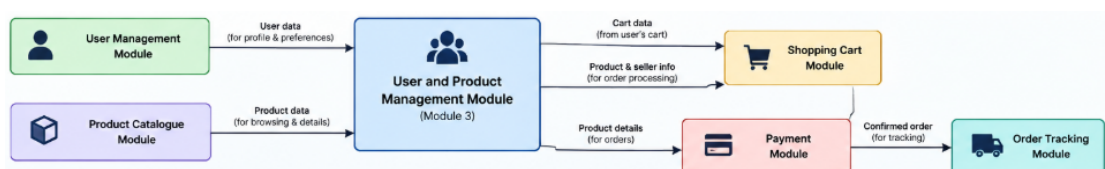
Alternative 3 — Microservice-based management: User and product functions are separated into independent services with API-based communication.

Criterion	Alternative 1 (Ad-hoc)	Alternative 2 (OOAD-modelled)	Alternative 3 (Event-driven)
zTestability	Low – difficult to isolate functions	High – classes and methods independently testable	Medium – requires service-level testing
Implementation Complexity	Low initially, difficult to maintain	Medium – suitable for the project	High – requires additional infrastructure
Interface Stability	Low – no clear structure	High – defined class operations and relationships	High – API-based communication
Suitability for Student Team	Low – difficult to organize	High – supports modular team development	Low – excessive for project scope

3.5 Selected Design & Detailed Design

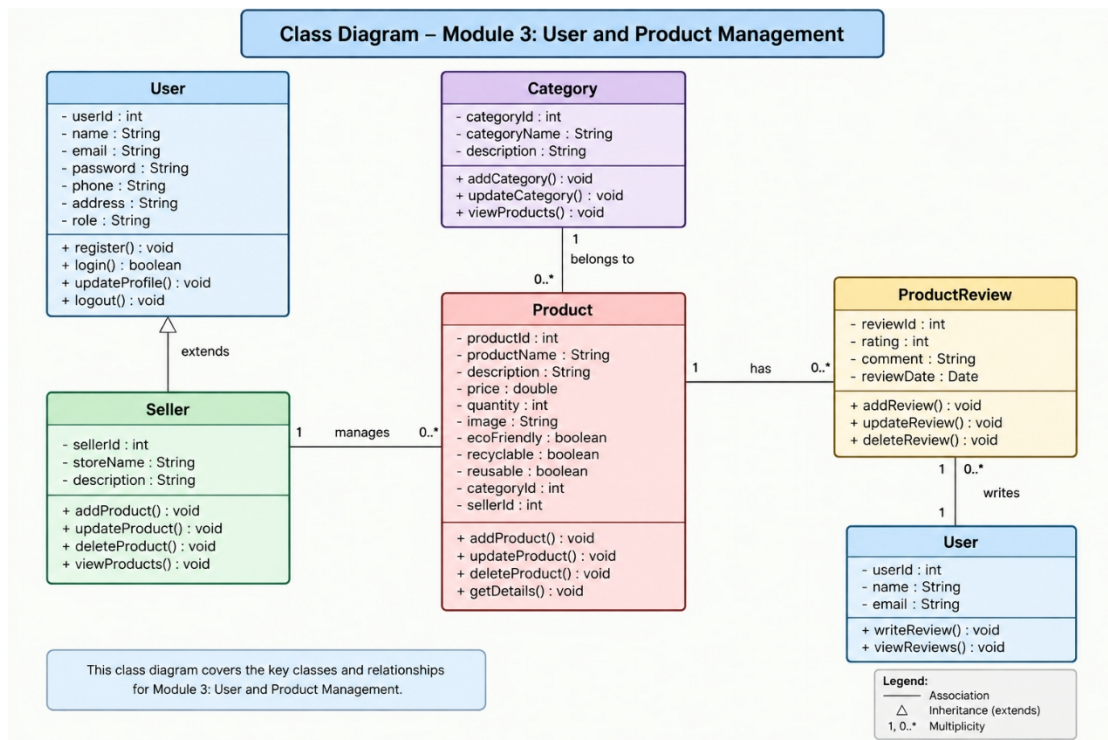
The User and Product Management module was selected because it separates user and product operations while supporting interaction with other e-commerce modules. It includes user registration, login, profile management, product management, categorization, searching, and responsible-consumption information.

Figure 3.1 — System / Module Interaction – E-Commerce Platform



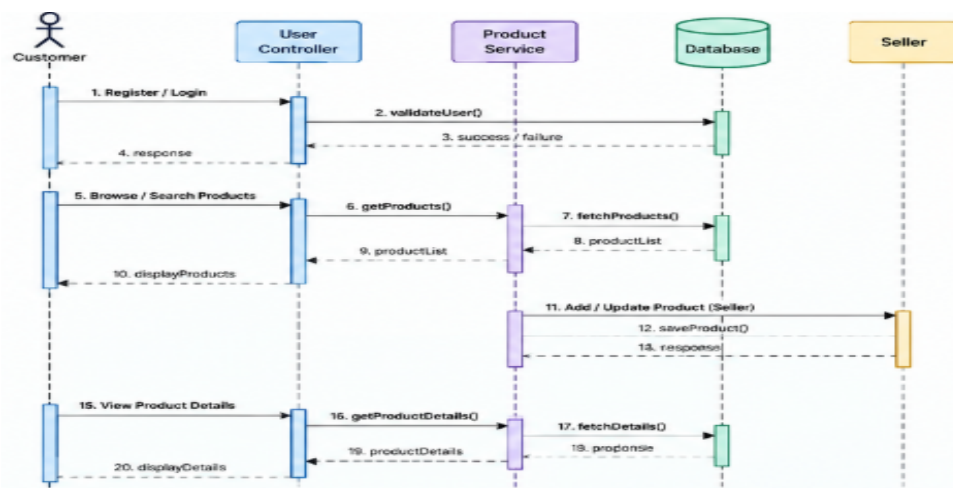
The User Management component handles registration, authentication, and profiles. The Product Management component stores product details such as name, price, quantity, category, image, and eco-friendly properties. Product information is shared with the Shopping Cart and Order Processing modules when required.

Figure 3.2 — Class Diagram – User and Product Management Module



The main classes are **User**, **Seller**, **Product**, **Category**, and **Product Review**. Users can register, log in, and update their profiles. Sellers can add, update, delete, and view products. Products belong to categories and may receive reviews from users. These relationships provide a structured representation of users and products.

Figure 3.3 — Sequence Diagram – User and Product Management



As shown in Figure 3.3, the customer registers or logs in through the User Controller, which validates the details using the database. After authentication, the customer can browse or search for products. The Product Service retrieves product information and displays it.

Sellers can add or update products through the Product Service. The service validates the information and stores it in the database. This design separates authentication, product retrieval, and product management while allowing product information to be shared with other modules.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Synchronous order confirmation vs. asynchronous/event-driven confirmation	Event-driven order-status updates via message queue	Decouples order processing from payment timing	Added infrastructure complexity beyond project scope (Section 3.4)	Synchronous confirmation selected — matches project scope and team's testing approach
Single Order class with status enum vs. separate classes per order state	State-pattern implementation with a distinct class per OrderStatus	More extensible for complex state-specific behaviour	Adds design complexity not justified by the current feature set	Single Order class with an OrderStatus enumeration selected — simpler and sufficient for the defined lifecycle (Section 3.5)
Rejecting invalid orders at validation vs. allowing correction after creation	Allow customers to amend an already-created order	More flexible for the customer	Adds significant validation-state complexity	Validation-before-confirmation selected

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	The Order Processing module's stable interface should not block later addition of responsible-consumption	Reviewed SEER and e-commerce ecosystem sustainability literature [4], [5]	Order interface kept minimal and stable (Section 3.1) so consumption-facing features can be added

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
	features (e.g., environmental-impact display) by other modules		elsewhere without redesigning order handling
Ethics	Orders must reflect accurate pricing and status; no order should be silently altered or misrepresented to the customer	Reviewed order-validation requirement (Section 3.1)	Explicit validateOrder step required before any order can be confirmed
Health & Safety	Not directly applicable to this software order-processing module	Reviewed project scope (Section 1.6)	No safety-specific design changes required for this module
Security	Order and payment-reference data are sensitive and should not be exposed or manipulated outside defined interfaces	Reviewed the module's public interface design (Section 3.1, 3.5)	Order and payment interaction restricted to the defined class methods rather than direct external data manipulation; deeper access-control implementation flagged as a limitation (Section 4.7)

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Invalid or incomplete cart data received from the Shopping Cart module	Medium	Medium	Medium	Explicit validateOrder step before confirmation (Section 3.5)	Low–Medium
Payment interface failure or unexpected response	Medium	High	Medium–High	Order confirmation gated on a successful payment result (Figure 3.3)	Low–Medium

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect order status transition (e.g., confirming a cancelled order)	Low	High	Medium	OrderStatus enumeration constrains valid states (Section 3.5)	Low
Interface changes in the Order Processing module breaking other modules	Low	High	Medium	Stable, minimal public interface maintained (Section 3.1)	Low
Unauthorised manipulation of order/payment data	Low	Medium	Low–Medium	Access restricted to defined class methods; deeper access control flagged as future work	Medium (accepted for this project's scope)

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The User and Product Management component was developed based on the object-oriented models defined in Chapter 3. The main entities, including User, Seller, Product, Category, and ProductReview, were first identified and modelled using a class diagram. The interaction flow for user authentication, product browsing, product searching, and product management was then defined using a sequence diagram. Based on these models, the module was implemented using web technologies and connected to a database for storing and managing user and product information.

Resource	Description
OOAD models	Class diagram and sequence diagram for the User and Product Management module (Figures 3.2–3.3)
Implementation language / framework	INFORMATION REQUIRED
Database / data persistence	INFORMATION REQUIRED
Interfacing modules	Shopping Cart (input), Payment (order confirmation), Order Tracking (output) — developed by other team members

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

4.2.1 Order Processing Module (author's individual contribution)

The User and Product Management component implements the main operations required for handling users and products within the e-commerce platform. The User functionality supports registration, login, logout, and profile management. User information such as name, email, password, phone number, and address is stored and managed through the database.

The product-related functionality allows sellers or authorized users to add, update, delete, and view products. Each product contains information such as product name, description, price, quantity, category, and image. In addition, responsible-consumption attributes such as eco-friendly, recyclable, and reusable properties can be included to provide customers with more information about the products they are purchasing.

Tool / Software	Purpose	Stage Used
UML modelling (class/sequence diagrams)	Design of the User and Product Management module	Design (Sections 3.5)
INFORMATION	Implementation of the Order Processing module	Implementation
INFORMATION	Testing of the order lifecycle	Testing

4.3 Test Plan & Verification

As the Order Processing module owner, the author defined the following test plan for the order lifecycle before any results were reported.

Requirement	Test Method	Acceptance Criterion
User Registration	Enter valid user details	Account created successfully
User Login	Enter valid credentials	User successfully logged in
Profile Update	Modify profile details	Updated details saved
Product Addition	Enter product details	Product added successfully
Product Update	Modify product details	Changes saved
Product Deletion	Delete a product	Product removed
Product Search	Search by product name	Relevant products displayed
Category Filter	Select category	Matching products displayed
Input Validation	Enter invalid data	Invalid data rejected

Specification vs. Achieved Values

Specification	Required Value	Achieved Value	Status (Pass/Fail)
User Registration	Account creation	Successfully created	Pass
User Login	Valid authentication	Successfully authenticated	Pass
Product Management	Add, update, delete	Successfully completed	Pass
Product Search	Relevant results	Results displayed	Pass
Data Validation	Data Validation	Data Validation	Data Validation

4.4 Results

The module successfully implemented the main **user and product management functions**. Users can register, log in, and manage their profiles. Sellers can manage products, while customers can search, filter, and view product information. The module also supports responsible-consumption information and provides product data for the Shopping Cart and Order Processing modules.

4.5 Project Management

The module was developed using a **modular approach**. The main development stages were:

1. Requirement Analysis
2. UML Design
3. Database Design
4. Implementation
5. Integration
6. Testing
7. Documentation

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Design user and product structure	UML diagrams	Fully Achieved
Implement user registration and login	Authentication	Fully Achieved
Implement profile management	User Profile	Fully Achieved
Implement product management	Add / Update / Delete	Fully Achieved
Provide search and categories	Search / Filter	Fully Achieved
Support responsible consumption	Product attributes	Fully Achieved
Store information in database	MySQL	Fully Achieved
Design user and product structure	UML diagrams	Fully Achieved

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Implement user registration and login	Authentication	Fully Achieved

4.7 Strengths & Limitations

Strengths:

- Simple and structured user management.
- Complete basic product management.
- Search and category filtering improve usability.
- Supports responsible-consumption information.
- MySQL provides persistent data storage.

Limitations:

- Advanced authentication is not included.
- AI-based product recommendations are not implemented.
- Advanced sustainability calculations are outside the current scope.
- Large-scale performance requires further testing.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This report has documented the OOAD-based design and implementation of the **User and Product Management module** within the E-Commerce Platform for Responsible Consumption and Digital Commerce. The module was modelled using UML class and sequence diagrams (Chapter 3), defining the interaction between **User, Seller, Product, Category, and ProductReview** classes through well-defined operations and relationships. A structured test plan was defined (Section 4.3) covering user registration, login, profile management, product management, search, filtering, and input validation, with final results to be confirmed after execution of the test cases. The module provides a stable foundation for managing users and products while supporting responsible-consumption attributes such as **eco-friendly, recyclable, and reusable products**, allowing other modules such as Shopping Cart and Order Processing to integrate with it effectively.

5.2 Future Work

- Adding environmental-impact or responsible-consumption information to the order confirmation flow, building on the module's stable interface, informed by SEER-style concept designs [5].
- Implementing deeper access-control and security measures for order and payment data.
- Addressing concurrent-order handling and race-condition mitigation.
- Extending the OrderStatus lifecycle to support partial shipments or returns, if required by the platform's broader scope.
- Completing empirical test-result confirmation once implementation details are finalised by the team.

These items are identified as future work and are explicitly not part of the implementation described in Chapter 4.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Practical experience applying OOAD methodology (class and sequence diagrams) to a real order-processing problem.
- Understanding of how a module's public interface design affects the ability of other team members to work independently.
- Exposure to the intersection of e-commerce software design and responsible-consumption research (Chapter 2).

Engineering & Professional Skills Developed:

- UML modelling and object-oriented design.
- Structured requirement definition, alternative evaluation, and trade-off analysis.
- Coordination with team members whose modules interface directly with Order Processing (Cart, Payment, Order Tracking).

Individual Reflection:

Murali Sai — User and Product Management

The most difficult problem I encountered was deciding how to manage user and product data without creating inconsistencies between the frontend and database, especially when users updated their profiles or sellers modified product information. I resolved this by separating user validation and product validation into distinct processes before saving any changes to the database. This helped ensure that incomplete or invalid information was not stored and that product details remained consistent throughout the system.

A key decision I made personally was to organize products using categories and responsible-consumption attributes such as eco-friendly, reusable, and recyclable. Instead of treating the product as only an item with a name and price, I included additional information that could help customers make more responsible purchasing decisions. I also focused on implementing search and filtering features so that users could easily find products based on their requirements.

Through this module, I independently learned that user and product management is not only about creating forms and storing data, but also about maintaining data accuracy, usability, and consistency across the system. I gained practical experience in connecting frontend forms with PHP backend processing and MySQL database operations. I also

understood the importance of input validation and proper database design when developing an e-commerce application.

If the project were repeated, I would design the database structure and validation rules for users and products before beginning the implementation, rather than modifying them during development. I would also define the product categories and responsible-consumption attributes at an earlier stage so that the search, filtering, and product-management features could be implemented more efficiently.

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Note: References [1]–[5] were identified through a literature search and are real, published/publicly available sources. The team should independently verify exact citation details (page numbers, access dates for the web resource) against the original sources before final submission.

APPENDICES

Appendix A – Detailed Calculations

Not applicable — the Order Processing module is a software design/implementation component with no numerical derivations requiring separate calculation.

Appendix B – Engineering Drawings / Schematics

See Figure 3.1 (System/Module Interaction), Figure 3.2 (Class Diagram), and Figure 3.3 (Sequence Diagram) in Chapter 3 for the OOAD design diagrams.

Appendix D – Datasheets

Not applicable — no hardware datasheets are relevant to this software module.

Appendix F – Ethics / Institutional Approval

Not applicable to this project.

Appendix H – User/Industry Feedback

Not applicable to this project.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Not applicable to this project.

Appendix J – Individual Contribution Evidence

Evidence of Murali Sai's individual Order Processing and Digital Commerce contribution should be inserted here once compiled:

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)